Lecture 11

11 February 2026 19:57

Recap

Ex. ① $f: \mathbb{R}^n \longrightarrow \mathbb{R}$, $f(x) = \|x\|$

f is strictly convex.

Here $f(\alpha x + (1-\alpha) y) \leq \alpha f(x) + (1-\alpha) f(y)$

$\|\alpha x + (1-\alpha) y\| \leq \alpha \|x\| + (1-\alpha) \|y\|$

② $g: \mathbb{R}^n \longrightarrow \mathbb{R}$

$g(x) = \|x - y\|$ where y is fixed.

we can show that

$$\underbrace{\|\alpha x_1 + (1-\alpha) x_2 - y\|}_{g(\alpha x_1 + (1-\alpha) x_2)} \leq \alpha \|x_1 - y\| + (1-\alpha) \|x_2 - y\|$$

$\alpha g(x_1) + (1-\alpha) g(x_2)$

- using triangle inequality

$g(x)$ is convex for x (for any y)

③ $h: \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$

$h(x, y) = \|x - y\|$ ✓

To claim that h is a convex fn, we must show that for all $\alpha \in [0,1]$ and $(x_1, y_1), (x_2, y_2)$,

$$h(\alpha (x_1, y_1) + (1-\alpha)(x_2, y_2)) \leq \alpha h(x_1, y_1) + (1-\alpha) h(x_2, y_2).$$

$(x_2, y_2) \in \mathbb{R}^{2n}$

$(x_1, y_1) \in \mathbb{R}^{2n}$

On other words,

$$h(\alpha x_1 + (1-\alpha) x_2, \alpha y_1 + (1-\alpha) y_2) \leq \alpha h(x_1, y_1) + (1-\alpha) h(x_2, y_2)$$

or,

$$\|\alpha x_1 + (1-\alpha) x_2 - (\alpha y_1 + (1-\alpha) y_2)\| \leq \alpha \|x_1 - y_1\| + (1-\alpha) \|x_2 - y_2\|$$

- this can also be shown triangle inequality. (Exercise)

To summarise $h: \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}$

we can have $h(x, y)$ a convex fn. —— ①

or, $h(x, y)$ convex in x for each given y —— (II)

or $h(x, y)$ convex in y for each given x —— (III)

① $\Longrightarrow$ (II) and (III)

(II) or (III) $\not\Longrightarrow$ ①

① $\Longleftrightarrow$ $h(\alpha x_1 + (1-\alpha) x_2, \alpha y_1 + (1-\alpha) y_2) \leq \alpha h(x_1, y_1) + (1-\alpha) h(x_2, y_2)$

this implies

(II) $h(\alpha x_1 + (1-\alpha) x_2, y) \leq \alpha h(x_1, y) + (1-\alpha) h(x_2, y)$ for all y

for all x

(iii) and $h(x, \alpha y_1 + (1-\alpha) y_2) \le \alpha h(x, y_1) + (1-\alpha) h(x, y_2)$ [illegible]

Maximum of convex functions ($S \subseteq \mathbb{R}^n$ convex, $f_i: S \to \mathbb{R}$, $i \ge 1$):
Define $g, h: S \to \mathbb{R}$ as

$$g(x) = \max_i f_i(x)$$

$$h(x) = \min_i f_i(x)$$

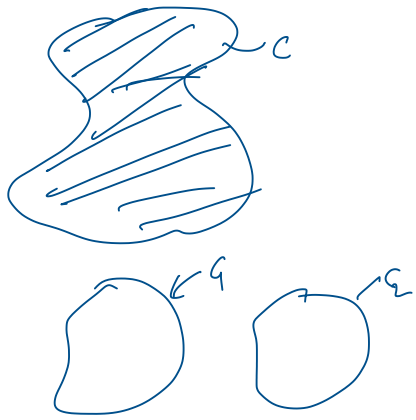
- If $f_i, i \ge 1$ are ~~convex functions~~ so is g.
- If $f_i, i \ge 1$ are concave functions so is h.

Example $h(x, y) = \|x - y\|$ convex in x for any y

define $f: \mathbb{R}^n \to \mathbb{R}$ as

$$f(x) = \max_{y \in C} \|x - y\|$$

$\Rightarrow f$ is also convex.



$C = C_1 \cup C_2$

Minimum of convex functions ($S_1 \subseteq \mathbb{R}^m, S_2 \subseteq \mathbb{R}^n$ convex, $f: S_1 \times S_2 \to \mathbb{R}$): Define $g: S \to \mathbb{R}$ as

$$g(x) = \inf_y f(x, y)$$

- If f ~~is a convex function~~ so is g.

Example $f: \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$

$f(x, y) = \|x - y\|$

define $g: \mathbb{R}^n \to \mathbb{R}$

$$g(x) = \min_{y \in C} \|x - y\|$$

then g is also convex.

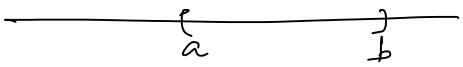
$f(x, y) = x^2 \sqrt{y}$, $y \ge 0$

$g(x) = \min_{y \in \mathbb{R}_+} f(x, y) = 0 \quad \forall x$

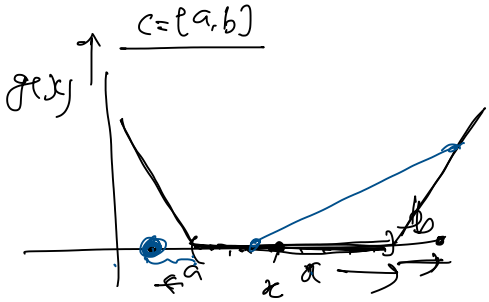
$f: \mathbb{R} \times \mathbb{R} \to \mathbb{R}$

$f(x, y) = |x - y|$

$$g(x) = \min_{y \in [a, b]} |x - y|$$



$C = [a, b]$



for $x \le a$

$g(x) = a - x$

for $x \ge b$

$g(x) = x - b$

$g(\alpha x_1 + (1-\alpha) x_2) \le \alpha g(x_1) + (1-\alpha) g(x_2)$

Example $f(x,y) = (x^2-x)\sqrt{y}$ $\quad 1 \le y \le 4$

For f to be convex in (x,y)

$$f(\alpha x_1 + (1-\alpha)x_2,\ \alpha y_1 + (1-\alpha)y_2) \le \alpha f(x_1,y_1) + (1-\alpha) f(x_2,y_2)$$

which doesn't hold

but f is convex in x for any fixed y.

$f : [0,2] \times [1,4] \longrightarrow \mathbb{R}$

convex in x for any given y

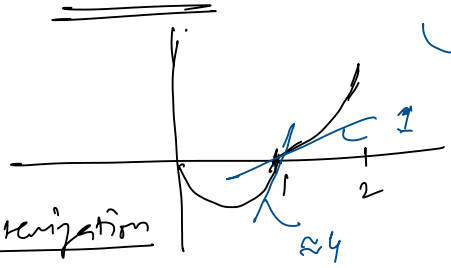
define $h(x) = \min_{y \in [1,4]} (x^2-x)\sqrt{y}$

$$= \begin{cases} 4(x^2-x) & x \in [0,1) \\ (x^2-x) & x \in [1,2] \end{cases}$$

$x^2 - x < 0 \iff x < 1$

(not convex)

$$h'(x) = \begin{cases} 4(2x-1) & x < 1 \\ (2x-1) & x > 1 \end{cases}$$



For convex functions

second order characterization

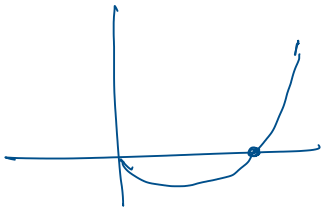
$f''(x) \ge 0 \Rightarrow$

i.e. $f'(x)$ should be increasing in x.

$g(x) = \max_{y \in [1,4]} (x^2-x)\sqrt{y}$

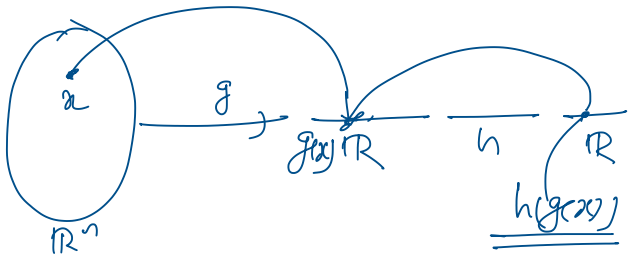
$\Rightarrow g$ is convex

$$g(x) = \begin{cases} (x^2-x) & x \in [0,1) \\ 4(x^2-x) & x \in [1,2] \end{cases}$$



Composition of functions

$h : \mathbb{R} \to \mathbb{R}, \quad g : \mathbb{R}^n \to \mathbb{R}$



define $f : \mathbb{R}^n \longrightarrow \mathbb{R}$

$f(x) = h(g(x))$

outer fn $\quad$ inner fn

h, g are twice differentiable

Assume $h : \mathbb{R} \longrightarrow \mathbb{R}, \quad g : \mathbb{R} \longrightarrow \mathbb{R}$

$f'(x) = h'(g(x))\, g'(x)$

$f''(x) = \underbrace{h''(g(x))}_{\geq 0} \cdot \underbrace{(g'(x))^2}_{\geq 0} + \underbrace{h'(g(x))}_{\geq 0}\, \underbrace{g''(x)}_{\geq 0}$

suppose h is convex and nondecreasing and g is convex

$\Rightarrow f$ is convex.

Composition of functions ($h: \mathbb{R} \to \mathbb{R}, g: \mathbb{R}^n \to \mathbb{R}$): Define $f: \mathbb{R}^n \to \mathbb{R}$ as

$$f(x) = h(g(x))$$

h	g	f
Convex and nondecreasing	Convex	Convex
Convex and nonincreasing	Concave	Convex
Concave and nondecreasing	Concave	Concave
Concave and nonincreasing	Convex	Concave

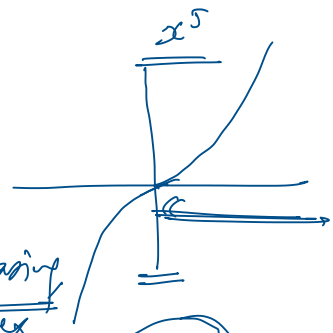
Examples ① $f(x) = \|x\|^5$

$g(x) = \|x\|$ — convex

$h: \mathbb{R}_+ \to \mathbb{R}$ $\quad h(x) = x^5$ — nondecreasing & convex

$f(x) = h(g(x))$

$\Rightarrow f$ is convex



$\|x\|^4$

$\|x\|$ convex

x^4 convex and nondecreasing for $x \geq 0$

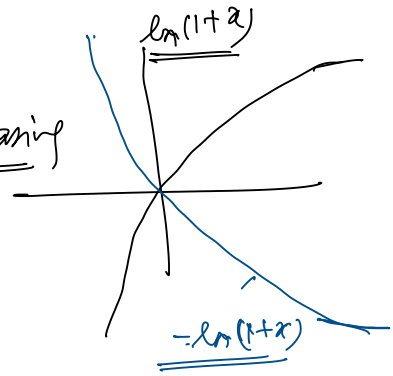
② $f(x) = -\ln(1+\sqrt{x})$

$g(x) = \sqrt{x}$ — concave ✓

$h(x) = -\ln(1+x)$ — convex, nonincreasing

$f(x) = h(g(x))$

$\Rightarrow f$ is convex.



③ $f(x) = -\ln(1+x^2)$

$g(x) = x^2$ — convex

$h(x) = -\ln(1+x)$ — convex and nonincreasing

- we cannot apply the above properties

$f: \mathbb{R} \to \mathbb{R}$, differentiable

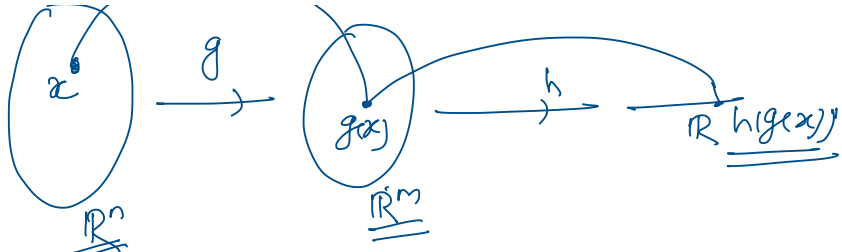
f nondecreasing, $f'(x) \geq 0$

f nonincreasing, $f'(x) \leq 0$

$f(x) = -\ln(1+x)$

$f'(x) = -\frac{1}{1+x}$

$\Rightarrow f'(x) \leq 0 \quad \forall x \geq 0$



$f: \mathbb{R}^n \to \mathbb{R}$

$f(x) = h(g(x))$

Recall the first property

h convex and non-decreasing g convex

g is convex $\iff$

$g(\alpha x + (1-\alpha) y) \le \alpha g(x) + (1-\alpha) g(y)$

ith component of g

i.e. $g_i(\alpha x + (1-\alpha) y) \le \alpha g_i(x) + (1-\alpha) g_i(y) \quad \forall i = 1, \dots, m$

i.e. g_i's are convex $\forall i = 1, \dots, m$

we call $s: \mathbb{R} \to \mathbb{R}$ nondecreasing if

$s(y) \ge s(x)$ $y \ge x$ $x, y \in \mathbb{R}$

$h: \mathbb{R}^m \to \mathbb{R}$

h is called nondecreasing in ith component if

$h(x_1, \dots, x_{i-1}, y_i, x_{i+1}, \dots, x_n) \ge h(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_n)$ if $y_i \ge x_i$

$\forall x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n$

$\left(\text{i.e. } \frac{\partial h(x)}{\partial x_i} \ge 0 \right)$.

(h is componentwise non-decreasing $\iff$ h is non-decreasing in all the components)

Composition of functions ($h: \mathbb{R}^m \to \mathbb{R}$, $g: \mathbb{R}^n \to \mathbb{R}^m$): Define $f: \mathbb{R}^n \to \mathbb{R}$ as

$$f(x) = h(g(x))$$

h	g_is	f
Convex and component-wise nondecreasing	Convex	Convex
Convex and component-wise nonincreasing	Concave	Convex
Concave and component-wise nondecreasing	Concave	Concave
Concave and component-wise nonincreasing	Convex	Concave

$\min_{x \in S} f(x)$

– local minima

– global minima

– 1st order / 2nd order necessary / sufficient conditions

Minima of convex functions ($S \subseteq \mathbb{R}^n$ convex, $f: S \to \mathbb{R}$ convex):

$$\min_{x \in S} f(x)$$

- If x^* is a local minimum then it also a global minimum.

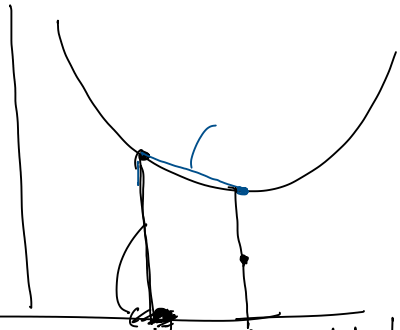
- If f ~~is strictly convex~~ then there is a ~~unique minimum~~.
- (S ~~open~~) x^* is a minimum $\Leftrightarrow$ ~~$\nabla f(x^*) = 0$~~. (necessary as well as sufficient condition)

Proof ① x^* is global min

$\bar{x} \neq x^*$ s.t.

$f(\bar{x}) > f(x^*)$

(If $f(\bar{x}) = f(x^*) \Rightarrow \bar{x}$ is also a global min.)



claim if f is convex then $\bar{x}$ cannot be a local minimum

$\bar{x}$ x^* global minimum

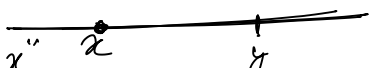
$\forall \alpha \in (0,1)$

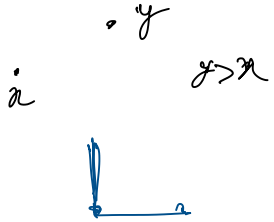
$f(\alpha \bar{x} + (1-\alpha) x^*) \leq \alpha f(\bar{x}) + (1-\alpha) f(x^*)$

$< f(\bar{x})$

$\Rightarrow \bar{x}$ cannot be a local minimum.

$x, y \in \mathbb{R}$



$f(y) \geq f(x)$ if "$y > x$"

$y > x \Leftrightarrow y - x > 0$